Fast integrators with a vertically implicit step on a terrain-following grid: Verlet EI and the WRF forward–backward scheme

MultiRate.jl

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Abstract

We formulate the linearly implicit Verlet step "Version EI" of the note *Split-explicit with implicit treatment of part of the fast waves* (Imex.pdf, 2024) for the fast part of the Gal-Chen finite-volume discretization, derive the tridiagonal system for the vertical momentum W, discuss when the contravariant vertical mass flux Ω has to be the unknown instead, and compare the scheme with the forward–backward acoustic step of WRF with its vertically implicit, off-centred Crank–Nicolson step. Both are implemented as fast integrators of the frozen MIS split with the W system and an explicit metric term (Section 8) and compared on the Agnesi mountain wave (Section 9).

Contents

1 Introduction **2**

2 The fast system on the terrain-following grid **2**

3 Verlet EI **3**
3.1 The step . . . 3
3.2 The system for W . . . 3
3.3 Properties . . . 3

4 W or the contravariant Ω? **4**
4.1 Metric term explicit: the two are equivalent . . . 4
4.2 The price of the explicit metric term . . . 4
4.3 Metric term implicit: the system for Ω . . . 4

5 The WRF forward–backward scheme **4**
5.1 WRF . . . 4
5.2 The analogue on the Gal-Chen grid . . . 5

6 Linear algebra **5**
6.1 The column system . . . 5
6.2 The steps as compositions . . . 6
6.3 The options in both variants: off-centring and extrapolation . . . 7
6.4 Fourier analysis . . . 8

7 Comparison **9**
7.1 Horizontal: the same spectrum . . . 9
7.2 Vertical: damping . . . 9
7.3 Summary . . . 10

8 Implementation in the frozen split **10**

9 Agnesi mountain wave **11**

10 Open decisions **14**

1 Introduction

Split-explicit methods (MIS, RK3) integrate the fast acoustic–gravity part with small steps $\Delta\tau$. On grids with $\Delta z \ll \Delta x$ the vertical acoustic Courant number $c_s\Delta\tau/\Delta z$ is far above one, and the vertical part has to be implicit, while the horizontal part stays explicit: HEVI. Three ingredients decide the properties of such a small step:

1. the explicit horizontal integrator (Störmer–Verlet, forward–backward);
2. the implicit vertical integrator (Crank–Nicolson, off-centred) and the variable whose column system is solved (W, Ω);
3. the place of the terrain metric terms (explicit or implicit).

The scheme of the Imex note combines the first two in one Verlet step (Section 3); WRF (Klemp, Skamarock and Dudhia 2007; Skamarock et al. 2019) uses a forward–backward step with an off-centred vertically implicit part (Section 5). The comparison is in Section 7.

2 The fast system on the terrain-following grid

Gal-Chen coordinate $z = z_s(x) + \eta J(x)$, $J = 1 - z_s/H$, C-grid with ρ, Θ (and π) at the cell centres, $U = \rho u$ at the x-faces, $W = \rho w$ at the η-faces; $\Omega = W - z_x\bar{U}$ is the contravariant mass flux through the η-surfaces ($\bar{U}$ the four-point mean; $\Omega = 0$ at the ground and the top). Thermodynamic variables $t = (\rho, \Theta)$, momenta (U, W). The fast part is

$$\begin{aligned}
\rho_t &= -\tfrac{1}{J}\big[\delta_x(J_x U) + \delta_\eta \Omega\big],\\
\Theta_t &= -\tfrac{1}{J}\big[\delta_x(\theta_x J_x U) + \delta_\eta(\theta_\eta \Omega)\big],\\
U_t &= -\tfrac{1}{J_x}\big[\delta_x(Jp) - \delta_\eta(z_x\, p_\sigma)\big],\\
W_t &= -\frac{\delta_\eta p}{J\Delta\eta} - g\bar{\rho},
\end{aligned}$$

p_σ the mean at the η-faces, θ_x, θ_η frozen face values, and $p = p(\Theta)$ (full pressure), $p \approx P\,\Theta$ with frozen $P = (\gamma p/\Theta)_f$ (product form) or the Exner pressure as an additional variable. We use the blocks

$$t_t = D_h U + D_v W + f_t, \qquad U_t = G_x\, p + f_U, \qquad W_t = G_z\, p - g\bar{\rho} + f_W,$$

D_h the horizontal divergence *including* the part $-z_x\bar{U}$ of Ω, D_v the part with W, G_x the horizontal pressure gradient *including* the metric term $\delta_\eta(z_x p_\sigma)/J_x$, G_z the vertical one; f the constant forcing of the stage. Two observations shape everything that follows:

- a vertical pressure difference acts on W *and*, through the metric term, on U, and U feeds back into the vertical divergence through Ω;
- for $z_x = 0$ the vertical part (D_v, G_z, g) is a closed column system.

3 Verlet EI

3.1 The step

In the Imex note the momentum operator is split into an explicit part M^E_{up} and an implicit part $M^I_{up}, M^I_{u\rho}$; the divergence M_{pu} is not split. Here $M_{pu} = D_h + D_v$, $M^E_{up} = G_x$ (U rows), $M^I_{up} = G_z$, $M^I_{u\rho} = -g\,\overline{(\cdot)}$ (W rows). With $h = \Delta\tau$ and the off-centring γ (Version III; $\gamma = \frac{1}{2}$ is Version EI):

$$t^{n+1/2} = t^n + \tfrac{h}{2}\big[D_h U^n + D_v W^n + f_t\big], \tag{1}$$

$$U^{n+1} = U^n + h\big[G_x\, p(t^{n+1/2}) + f_U\big], \tag{2}$$

$$W^{n+1} = W^n + h\big[(1-\gamma)(G_z p^n - g\bar\rho^n) + \gamma(G_z p^{n+1} - g\bar\rho^{n+1})\big] + h f_W, \tag{3}$$

$$t^{n+1} = t^{n+1/2} + \tfrac{h}{2}\big[D_h U^{n+1} + D_v W^{n+1} + f_t\big]. \tag{4}$$

The thermodynamic half steps are explicit in the full divergence; only W is implicit.

3.2 The system for W

U^{n+1} is known after (2). With $t^* = t^{n+1/2} + \frac{h}{2}[D_h U^{n+1} + f_t]$, (4) reads $t^{n+1} = t^* + \frac{h}{2} D_v W^{n+1}$. Linearize $p^{n+1} = p(t^*) + P\,(\Theta^{n+1} - \Theta^*)$ (exact for the product form and the Exner form; for the full pressure linearly implicit with $P = \gamma p/\Theta$ at the start of the stage) and insert into (3):

$$\Big[I - \tfrac{\gamma h^2}{2}\big(G_z P\, D_v^\Theta - g\,\overline{D_v^\rho}\big)\Big] W^{n+1} = W^n + h(1-\gamma)\big(G_z p^n - g\bar\rho^n\big) + h\gamma\big(G_z p(t^*) - g\bar\rho^*\big) + h f_W.$$

With $D_v^\rho W = -\delta_\eta W/(J\Delta\eta)$, $D_v^\Theta W = -\delta_\eta(\theta_\eta W)/(J\Delta\eta)$ and the face pressure difference $\alpha_k\Theta_k - \beta_k\Theta_{k-1}$ ($\alpha = \beta = P_f$ for the product form) this is, at the face k of a column with $\Delta z = J\Delta\eta$,

$$W_{k-1}\big[-r\beta_k\theta_{k-1} + g_g\big] + W_k\big[1 + r\theta_k(\alpha_k + \beta_k)\big] + W_{k+1}\big[-r\alpha_k\theta_{k+1} - g_g\big] = b_k,$$

$$r = \frac{\gamma h^2}{2\Delta z^2}, \qquad g_g = \frac{\gamma h^2 g}{4\Delta z},$$

a tridiagonal system per column; ρ^{n+1}, Θ^{n+1} follow from $t^{n+1} = t^* + \frac{h}{2} D_v W^{n+1}$. Equivalently: with the row scaling $D = \mathrm{diag}(\frac{h}{2}$ on t, γh on $W)$ the steps (3)–(4) are one solve $(I - DJ_v)Y = DG_v$ of the vertical part, preceded by the explicit vertical increments $\frac{h}{2} D_v W^n$, $(1-\gamma)hG_v$ in (1); the W system depends on the product $\frac{h}{2}\cdot\gamma h$ only. The elimination in detail, for all forms, is in Section 6.1.

3.3 Properties

- For a purely vertical problem ($D_h = G_x = 0$): $t^{n+1} = t^n + \frac{h}{2} D_v(W^n + W^{n+1})$ and (3): with $\gamma = \frac{1}{2}$ this is the trapezoidal rule (Crank–Nicolson) of the column system, unconditionally stable and non-dissipative.
- For a purely horizontal problem it is Störmer–Verlet (position Verlet), symmetric, second order, stable for $|\omega h| \le 2$.
- The step is symmetric for $\gamma = \frac{1}{2}$ and of second order; for $\gamma \ne \frac{1}{2}$ of first order.
- One column solve and two evaluations of the horizontal thermodynamic part, one of the horizontal momentum part, two of the vertical part per small step. The present Strang composition (`:hevi`: horizontal Verlet half step, vertical Crank–Nicolson, horizontal half step) needs one solve, four horizontal thermodynamic and two horizontal momentum evaluations.

4 W or the contravariant Ω?

4.1 Metric term explicit: the two are equivalent

In (2) the metric term $\delta_\eta(z_x p_\sigma)/J_x$ is part of G_x and explicit. Then $\bar{U}^{n+1}$ is known before the column solve, and $\Omega^{n+1} = W^{n+1} - z_x\bar{U}^{n+1}$ is an affine shift: the system for Ω has the same matrix and the right-hand side shifted by the known $z_x\bar{U}^{n+1}$. Nothing is gained by solving for Ω.

4.2 The price of the explicit metric term

Constant slope $s = z_x$, no gravity, small amplitude, vertical structure only. The fast system contains

$$p_t = -c_s^2\rho_0\,\partial_z(W - sU), \qquad W_t = -\frac{\partial_z p}{\rho_0}, \qquad U_t = s\,\frac{\partial_z p}{\rho_0} \quad\Longrightarrow\quad \Omega_t = -(1+s^2)\frac{\partial_z p}{\rho_0},$$

the vertical acoustic mode of the terrain-following coordinate has the frequency $c_s k_z\sqrt{1+s^2}$, and the part s^2 of the restoring force passes through the explicit U. A heuristic bound for the explicit coupling, from the largest vertical wavenumber $k_z \approx 2/(J\Delta\eta)$, is

$$\frac{c_s\,\Delta\tau\,|z_x|}{J\Delta\eta} \lesssim 1.$$

(This is an estimate, not yet the stability analysis of the coupled step.) Agnesi case of the MIS study: $350\cdot 4.3\cdot 0.026/200 \approx 0.2$; steep terrain, $z_x = 0.5$, $\Delta z = 50\,\mathrm{m}$: ≈ 15. For steep terrain the metric term has to be implicit.

4.3 Metric term implicit: the system for Ω

Move the vertical pressure difference of the U equation into the implicit part,

$$M^I_{up} = \big(\delta_\eta(z_x p_\sigma)/J_x \text{ on } U,\ G_z \text{ on } W\big), \qquad M^E_{up} = -\delta_x(Jp)/J_x.$$

The vertical divergence depends on the momenta only through Ω ($\delta_\eta\Omega$, $\delta_\eta(\theta_\eta\Omega)$), and

$$\Omega^{n+1} = \Omega' + h\big[(1-\gamma)G_\Omega p^n + \gamma G_\Omega p^{n+1}\big], \qquad G_\Omega p = G_z p - z_x\,\overline{\tfrac{1}{J_x}\delta_\eta(z_x p_\sigma)} \;\approx\; -(1+z_x^2)\frac{\delta_\eta p}{J\Delta\eta},$$

Ω' containing all explicit contributions. This is a tridiagonal system for Ω with the metric factor $1+z_x^2$ on the vertical pressure gradient; U^{n+1} is completed afterwards with the known p^{n+1}. Exactly column-local it is only with two approximations:

(a) the four-point means ($\bar{U}$ in Ω, p_σ and the average to the W face in the metric term) are taken within the column in the implicit part; exactly they couple neighbouring columns;

(b) in (4) the horizontal divergence $D_h U^{n+1}$ uses U before its implicit metric increment. With the full M_{pu}, as in the Imex note ($\mathcal{M} = I - \frac{1}{4}\Delta\tau^2 M^I_{up}M_{pu}$), the system for (U, W) is two-dimensional.

With (b) the step is no longer exactly symmetric, and the energy argument of the C-grid has to be checked again.

5 The WRF forward–backward scheme

5.1 WRF

WRF integrates the acoustic modes in the small steps of its RK3 (Wicker and Skamarock 2002), in a terrain-following *hydrostatic-pressure (mass)* coordinate with column mass μ, geopotential ϕ and perturbation variables about the state of the large step (Klemp, Skamarock and Dudhia 2007; Skamarock et al. 2019). Schematically, one small step $\tau \to \tau + \Delta\tau$:

1. *horizontal momentum forward*: $U^{\tau+\Delta\tau} = U^{\tau} + \Delta\tau\,[\text{horizontal pressure gradient of } p^* + \text{metric term } \partial_\eta p\,\partial_x\phi + R_U]$, with the divergence damping $p^* = p^{\tau} + \gamma_d(p^{\tau} - p^{\tau-\Delta\tau})$, $\gamma_d = 0.1$; the metric term is explicit;

2. *column mass and* Ω *backward*: $\mu^{\tau+\Delta\tau}$ from the vertically integrated horizontal divergence of $U^{\tau+\Delta\tau}$, then $\Omega^{\tau+\Delta\tau}$ diagnostically from the continuity equation of the column — in the mass coordinate Ω is *not* an unknown of the implicit system;

3. $\Theta^{\tau+\Delta\tau}$ with the horizontal fluxes of $U^{\tau+\Delta\tau}$ and the vertical ones of $\Omega^{\tau+\Delta\tau}$;

4. *vertically implicit*: W and ϕ with the off-centred average $\overline{a}^{\tau} = \frac{1+\beta_d}{2}a^{\tau+\Delta\tau} + \frac{1-\beta_d}{2}a^{\tau}$, $\beta_d = 0.1$, in the vertical pressure gradient (pressure from the linearized equation of state in ϕ_η, Θ, μ), the buoyancy and the geopotential equation $\phi_t \propto gW - \Omega\phi_\eta$; eliminating ϕ gives a tridiagonal system for W.

In addition: an external-mode filter and, on the large step, the RK3 with $n_s/3$, $n_s/2$, n_s small steps. The vertical unknown is the Cartesian W; the terrain enters the implicit system only through the explicit metric term and the diagnostic Ω.

5.2 The analogue on the Gal-Chen grid

In height coordinates $\Omega = W - z_x\bar{U}$ is not diagnostic; it couples W to the thermodynamic variables. The analogue (`:wrf` in `TimeStepping/fast.jl`, linearized fast operator), with $\beta = (1+\beta_d)/2 = 0.55$:

$$U^{n+1} = U^n + h\big[G_x\,p(t^*) + f_U\big], \qquad t^* = t^n + \gamma_d(t^n - t^{n-1}),$$
$$\begin{pmatrix} t \\ W \end{pmatrix}^{n+1} = \begin{pmatrix} t \\ W \end{pmatrix}^{n} + h\begin{pmatrix} D_h U^{n+1} + f_t \\ f_W \end{pmatrix} + h\,A_v\Big[(1-\beta)\begin{pmatrix} t \\ W \end{pmatrix}^{n} + \beta\begin{pmatrix} t \\ W \end{pmatrix}^{n+1}\Big],$$

A_v the vertical part $(D_v; G_z, -g)$. Elimination gives the W system of Section 3 with $r = \beta^2h^2/\Delta z^2$, $g_g = \beta^2h^2g/(2\Delta z)$; the metric term is explicit (as in WRF).

6 Linear algebra

6.1 The column system

In a column i with $\Delta z = J_i\Delta\eta$, cells $k = 1,\dots,n_z$ and faces $k = 1,\dots,n_z+1$ ($W_1 = W_{n_z+1} = 0$), the vertical part G_v acts only on the thermodynamic variables t and on W; with frozen coefficients it is linear,

$$J_v = \begin{pmatrix} 0 & D_v \\ B_v & 0 \end{pmatrix} \quad \text{on} \quad \begin{pmatrix} t \\ W \end{pmatrix},$$
$$(D_vW)_{\phi,k} = -\frac{\phi_{k+1}W_{k+1} - \phi_kW_k}{\Delta z}, \qquad (B_vt)_k = -\frac{\delta p_k}{\Delta z} - \frac{g}{2}(\rho_{k-1} + \rho_k),$$

ϕ the frozen face value of the variable ($\phi = 1$ for ρ, θ_η for Θ, $q_{v,\eta}$, $q_{c,\eta}$ for the moist variables), and the face pressure difference

form	δp_k	coefficients
product	$\alpha_k\Theta_k - \beta_k\Theta_{k-1}$	$\alpha_k = \beta_k = (\partial p/\partial\Theta)_{f,k}$, frozen
`:gradp`	$p(\Theta_k) - p(\Theta_{k-1})$	linearized: $\alpha_k\Theta_k - \beta_k\Theta_{k-1}$, $\alpha_k = (\gamma p/\Theta)_k$, $\beta_k = (\gamma p/\Theta)_{k-1}$ at the stage start
Exner, moist	$P_k(\pi_k - \pi_{k-1})$	$P_k = c_p\Theta_{f,k}$, $(D_vW)_{\pi,k} = -(c^B_{k+1}W_{k+1} - c^T_kW_k)/\Delta z$

For the full pressure the matrix is the Jacobian at the stage start (linearly implicit); for the other forms it is exact.

Two-factor solve. All schemes need $Y = (I - DJ_v)^{-1}R$ with the row scaling $D = \operatorname{diag}(c_t I_t,\, c_w I_W)$:

$$\begin{pmatrix} I & -c_t D_v \\ -c_w B_v & I \end{pmatrix} \begin{pmatrix} Y_t \\ Y_W \end{pmatrix} = \begin{pmatrix} R_t \\ R_W \end{pmatrix}.$$

The first block row gives $Y_t = R_t + c_t D_v Y_W$; inserted into the second (Schur complement on the W block):

$$\big(I - c_t c_w\, B_v D_v\big)\, Y_W = R_W + c_w\, B_v R_t.$$

With $r = c_t c_w/\Delta z^2$ and $g_g = c_t c_w g/(2\Delta z)$, row $k = 2, \dots, n_z$ of the product and `:gradp` forms is

$$\begin{aligned}\big[-r\beta_k\theta_{k-1} + g_g\big]Y_{W,k-1} + \big[1 + r\theta_k(\alpha_k + \beta_k)\big]Y_{W,k} + \big[-r\alpha_k\theta_{k+1} - g_g\big]Y_{W,k+1} \\ = R_{W,k} - \frac{c_w}{\Delta z}\big(\alpha_k R_{\Theta,k} - \beta_k R_{\Theta,k-1}\big) - \frac{c_w g}{2}\big(R_{\rho,k-1} + R_{\rho,k}\big),\end{aligned}$$

of the Exner and moist forms

$$\begin{aligned}\big[-rP_k c_{k-1}^T + g_g\big]Y_{W,k-1} + \big[1 + rP_k(c_k^T + c_k^B)\big]Y_{W,k} + \big[-rP_k c_{k+1}^B - g_g\big]Y_{W,k+1} \\ = R_{W,k} - \frac{c_w}{\Delta z}P_k\big(R_{\pi,k} - R_{\pi,k-1}\big) - \frac{c_w g}{2}\big(R_{\rho,k-1} + R_{\rho,k}\big)\end{aligned}$$

(the coefficient of $Y_{W,k-1}$ for $k = 2$ and of $Y_{W,k+1}$ for $k = n_z$ vanish). The pressure part is a discrete $-c_s^2\partial_{zz}$ with positive diagonal, the gravity part $\pm g_g$ is antisymmetric and small ($g_g/r = g\Delta z/2 \ll c_s^2$); the system is solved by the Thomas algorithm without pivoting, factored once per stage and value of $c_t c_w$ (`_frozen_factor`, `_exner_factor`, cached under the key $c_t c_w$). Back substitution:

$$Y_{\phi,k} = R_{\phi,k} - \frac{c_t}{\Delta z}\big(\phi_{k+1}Y_{W,k+1} - \phi_k Y_{W,k}\big) \quad \text{for every thermodynamic variable } \phi$$

(ρ, Θ, π with c^B, c^T, q_v, q_c); U rows: $Y_U = R_U$. Moisture enters B_v only through ρ and π, so the moist variables are back substitution only. The matrix depends on c_t and c_w only through the product; the schemes use

	c_t	c_w	$c_t c_w$
HEVI (Crank–Nicolson step of length h)	$h/2$	$h/2$	$h^2/4$
EI	$h/2$	γh	$\gamma h^2/2$
WRF	βh	βh	$\beta^2 h^2$

For $\gamma = \beta = \frac{1}{2}$ the three matrices are identical.

6.2 The steps as compositions

Let H be the linear (affine with the forcing) part "thermodynamics and vertical", acting on (t, W) with U as a parameter,

$$H(t, U, W) = \big(D_h U + D_v W + f_t,\ 0,\ B_v t + f_W\big),$$

and define for a row scaling $D = (c_t, c_w)$ the explicit and the implicit step of H and the momentum kick,

$$E_D\, z = z + D\, H(z), \qquad I_D\, z = z^+ \text{ with } z^+ = z + D\, H(z^+), \qquad K_h\, z = z + h\,(0,\, G_x p(t) + f_U,\, 0).$$

I_D is one two-factor solve: $z^+ = z + (I - DJ)^{-1}DH(z)$; U is constant in E_D and I_D, so the horizontal thermodynamic term $D_h U$ is the same explicit or implicit. Then (Sections 3, 5)

$$\boxed{\mathrm{EI}(\gamma) = I_{(h/2,\,\gamma h)} \circ K_h \circ E_{(h/2,\,(1-\gamma)h)}, \qquad \mathrm{WRF}(\beta) = I_{\beta h} \circ E_{(1-\beta)h} \circ K_h}$$

($I_{\beta h}E_{(1-\beta)h}$ is the β-weighted Crank–Nicolson step of H with the new U, the second line of the WRF step). Without H ($D_v = B_v = 0$) EI is position Verlet and WRF forward–backward; without K both are the (off-centred) Crank–Nicolson step of the column.

Equivalence for $\gamma = \beta = \frac{1}{2}$. With $c = h/2$ and $E_c = I + cH$, $I_c = (I - cH)^{-1}$ (affine: the constant parts commute as well, $(I+cH)(I-cH)^{-1}cf+cf = (I-cH)^{-1}2cf$), E_c and I_c commute, and

$$\mathrm{WRF}(\tfrac{1}{2}) = I_c E_c K_h = E_c I_c K_h = E_c \left(I_c K_h E_c\right) E_c^{-1} = E_c \,\mathrm{EI}(\tfrac{1}{2})\, E_c^{-1}.$$

This holds for arbitrary linear operators D_h, D_v, G_x, B_v — terrain (the $-z_x\bar{U}$ part of Ω and the metric term), gravity, sponge-free forcing — not only for Fourier modes. Consequences:

- the two small steps have the same eigenvalues: the same stability limit, the same dispersion and (none) dissipation of every mode;
- they differ by the change of variables E_c (an explicit half step of H): WRF's thermodynamic variables are EI's $t^{n+1/2}$, WRF's W is EI's $W + \frac{h}{2}B_v t$. Over a stage of n small steps

$$\mathrm{EI}^n = E_c^{-1}\,\mathrm{WRF}^n\,E_c = I_c K_h\,\mathrm{WRF}(\tfrac{1}{2})^{n-1}\,E_c;$$

- the combined half steps of Section 8 are exactly this formula: E_c at the stage start, then $n-1$ WRF steps ($E_c I_c K_h$; the update $Z \leftarrow 2Z - Z_s$ is E_c applied after I_c, since I_c gives $cH(z^+) = z^+ - z_s$), and the last step without E_c. EI with combined half steps *is* $\mathrm{WRF}(\frac{1}{2})$, started and finished correctly;
- in a MIS or RK3 stage t, U, W start at the same time level. EI integrates exactly the fast flow of these variables to second order; WRF integrates E_c-transformed variables but starts from untransformed ones and returns untransformed ones: the stage map differs from EI's by $\mathrm{WRF}^n - E_c^{-1}\mathrm{WRF}^n E_c = O(\Delta\tau)$ — the first-order "conjugation defect" of Section 7, and the only difference between the two schemes.

Verified in the code (frozen loop, hill grid, 10 steps of 2 s with constant forcing): $\|\mathrm{WRF}^n(E_c x) - E_c(\mathrm{EI}^n x)\|/\|x\| = 9\cdot10^{-16}$ (product), $3\cdot10^{-15}$ (Exner, moist), $2\cdot10^{-6}$ (`:gradp`, linearly implicit), while $\|\mathrm{WRF}^n x - \mathrm{EI}^n x\|/\|x\| = 4$–$8\cdot10^{-4}$.

6.3 The options in both variants: off-centring and extrapolation

Both variants have the same two optional parameters, default zero:

- the *off-centring* $\varepsilon \geq 0$ (WRF's `epssm`), $\beta = (1+\varepsilon)/2$;
- the *explicit extrapolation* $\gamma_d \geq 0$ (divergence damping, WRF's `smdiv`): the momentum kick uses the extrapolated thermodynamic state $t + \gamma_d(t - t_{\mathrm{prev}})$, t_{prev} the state of the previous kick,

$$K_h^{\gamma_d} z = z + h\left(0,\ G_x p(t + \gamma_d(t - t_{\mathrm{prev}})) + f_U,\ 0\right).$$

They enter the two schemes in the same places:

$$\boxed{\mathrm{WRF}(\varepsilon,\gamma_d) = I_{\beta h} \circ E_{(1-\beta)h} \circ K_h^{\gamma_d}, \qquad \mathrm{EI}(\varepsilon,\gamma_d) = I_{\beta h} \circ K_h^{\gamma_d} \circ E_{(1-\beta)h}}$$

with scalar weights on all rows of H (thermodynamic rows including $D_h U$ and f_t, W rows including f_W).

- *Off-centring.* In WRF, $I_{\beta h}E_{(1-\beta)h}$ is the β-weighted Crank–Nicolson step of the column with the new U: $(1-\beta)$ old and β new values of $D_v W$ in t and of $B_v t$ in W. In EI the first (explicit) part gets the weight $(1-\beta)h$ and the last (implicit) part βh, on all rows; $D_h U$ is constant within E and I, so only the split of the horizontal divergence between the old and the new U changes ($\frac{h}{2}, \frac{h}{2} \to (1-\beta)h, \beta h$). For $\varepsilon = 0$ EI is Verlet EI (Imex version 1).
- *Extrapolation.* In WRF the kick uses $\tau^n + \gamma_d(\tau^n - \tau^{n-1})$ with $\tau^{-1} = \tau^0$ (no extrapolation in the first step of a stage). In EI the kick acts on $t^{n+1/2}$, so the extrapolation is $t^{n+1/2} + \gamma_d(t^{n+1/2} - t^{n-1/2})$, started at the first $t^{1/2}$ (no extrapolation in the first step).

Conjugacy for all (ε, γ_d). $E_{(1-\beta)h} = I + (1-\beta)hH$ and $I_{\beta h} = (I - \beta h H)^{-1}$ are functions of the same H and commute; for the affine parts, with $Hz = Jz + a$,

$$E_{(1-\beta)h} I_{\beta h} z - I_{\beta h} E_{(1-\beta)h} z = \big[(I + (1-\beta)hJ)\beta h + (1-\beta)h(I - \beta h J) - h\, I\big](I - \beta h J)^{-1} a = 0.$$

Hence, with $E = E_{(1-\beta)h}$,

$$\mathrm{WRF}(\varepsilon, \gamma_d) = E\, \mathrm{EI}(\varepsilon, \gamma_d)\, E^{-1}, \qquad \mathrm{EI}^n = E^{-1} \mathrm{WRF}^n E = I_{\beta h} K_h\, \mathrm{WRF}^{n-1} E.$$

For the extrapolation the history has to correspond as well: if the WRF state is $w^n = Ez^n$ (z^n the EI state), WRF's τ^n is EI's $t^{n+1/2}$ for all n, and both extrapolations start without extrapolation; the kicks are identical. So for every choice of the two options EI and WRF have the same small step up to the change of variables E, the same eigenvalues, and the same stability; they differ only at the stage boundaries, where WRF starts and ends without E ($O(\Delta\tau)$ for $\varepsilon = 0$, $O((1-\beta)\Delta\tau)$ in general). For $\varepsilon > 0$ both are first order in $\Delta\tau$; for $\varepsilon = 0$ EI is second order and WRF first order (only through the stage boundaries).

Verified in the code (frozen loop, hill grid, 10 steps of 2 s, constant forcing, combined and separate half steps of EI): $\|\mathrm{WRF}^n(Ex) - E(\mathrm{EI}^n x)\|/\|x\| \le 3 \cdot 10^{-15}$ for $(\varepsilon, \gamma_d) = (0,0), (0.1,0), (0,0.1), (0.1,0.1), (0.2,0.3)$ in the product, Exner, pressure and moist forms; 2–$3 \cdot 10^{-6}$ for `:gradp` (linearly implicit).

The Imex version III (γ, W rows only). EI also keeps the off-centring of the Imex note, `EI_GAMMA[]` $= \gamma$, used only with $\varepsilon = 0$: $E_{(h/2,(1-\gamma)h)}$ and $I_{(h/2,\gamma h)}$ with different weights on the thermodynamic and the W rows. These do not commute for $\gamma \neq \frac{1}{2}$: version III is not conjugate to WRF for any β, and damps the vertical modes less (Fourier analysis below).

Combined half steps with the options. With the weights (e_t, e_w) of E and (i_t, i_w) of I (EI(ε): $e_t = e_w = (1-\beta)h$, $i_t = i_w = \beta h$; version III: $(\frac{h}{2}, (1-\gamma)h)$, $(\frac{h}{2}, \gamma h)$), I adds $z^+ - z_s = i_t H_t + D_{(i_t, i_w)} G_v(z^+)$ and the next E needs $e_t H_t + D_{(e_t, e_w)} G_v(z^+)$, so between two small steps

$$z \leftarrow z^+ + \frac{e_t}{i_t}(z^+ - z_s) + D_{(0,\ e_w - e_t i_w / i_t)} G_v(z^+),$$

the last term vanishing for EI(ε) and being $(1-2\gamma)hB_v(t^+)$ for version III; $\frac{e_t}{i_t} = \frac{1-\beta}{\beta}$ ($= 1$ for $\varepsilon = 0$: $2z^+ - z_s$). The extrapolation needs nothing extra.

6.4 Fourier analysis

For the acoustic model $t' = -\mathrm{i}(k_x u + k_z w)c_s$, $u' = -\mathrm{i}k_x c_s t$, $w' = -\mathrm{i}k_z c_s t$ with $a_x = k_x c_s h$, $a_z = k_z c_s h$, $\nu = (1-\gamma) + \gamma\lambda$, $\mu = (1-\beta) + \beta\lambda$, the ansatz $z^n = \lambda^n z$ gives, besides the neutral mode $\lambda = 1$,

$$\mathrm{EI}(\gamma):\ (\lambda-1)^2 + a_x^2\lambda + \tfrac{a_z^2}{2}\,\nu\,(1+\lambda) = 0, \qquad \mathrm{WRF}(\beta):\ (\lambda-1)^2 + a_x^2\lambda + a_z^2\,\mu^2 = 0.$$

(EI: from $t^{n+1/2}(\lambda - 1 + \frac{a_x^2}{2}) = t^n(\lambda - 1 - \frac{a_z^2}{2}\nu)$ and $t^n\lambda\big(\lambda - 1 + \frac{a_z^2}{2}\nu\big) = t^{n+1/2}\big(\lambda - 1 - \frac{a_x^2}{2}\lambda\big)$; the resultant factors as $(\lambda-1)[\ldots]$.) For $\gamma = \beta = \frac{1}{2}$, $\nu(1+\lambda)/2 = \mu^2 = (1+\lambda)^2/4$: identical polynomials, as proven above. Horizontally ($a_z = 0$) both are stable for $a_x \le 2$; vertically unconditionally. The modulus of the damped pair is the root of the ratio of constant and leading coefficient, $|\lambda|^2 = (1 + (1-\gamma)a_z^2/2)/(1 + \gamma a_z^2/2)$ (EI) and $(1 + (1-\beta)^2 a_z^2)/(1 + \beta^2 a_z^2)$ (WRF) for $a_x = 0$ — the formulas of Section 7. Spectral radius of the non-neutral modes:

a_x	a_z	EI $\gamma = 0.5$	EI 0.55	EI 0.6	WRF $\beta = 0.55$	WRF 0.5
0.0	1.0	1.0000	0.9802	0.9608	0.9608	1.0000
0.0	3.0	1.0000	0.9330	0.8699	0.8708	1.0000
0.0	20.0	1.0000	0.9054	0.8182	0.8198	1.0000
1.0	0.3	1.0000	0.9978	0.9956	0.9956	1.0000
1.5	3.0	1.0000	0.9330	0.8699	0.8708	1.0000
1.5	20.0	1.0000	0.9054	0.9512	0.8198	1.0000

EI with $\gamma \approx 0.6$ damps like WRF with $\beta = 0.55$, except for strongly coupled modes (large a_x and a_z), where EI's damping is weaker. On a grid (max over $a_z \in [0, 200]$) the horizontal limit is $a_x = 2$ for all variants.

7 Comparison

7.1 Horizontal: the same spectrum

For the oscillator $p' = -\omega u$, $u' = \omega p$, $a = \omega h$, forward–backward (momentum first) and Verlet (thermodynamic half step, momentum, half step) have the matrices

$$M_{FB} = \begin{pmatrix} 1 - a^2 & -a \\ a & 1 \end{pmatrix}, \qquad M_V = \begin{pmatrix} 1 - \frac{a^2}{2} & -a\left(1 - \frac{a^2}{4}\right) \\ a & 1 - \frac{a^2}{2} \end{pmatrix},$$

both with trace $2 - a^2$ and determinant 1: the same eigenvalues, stable for $|a| \leq 2$, no damping. They are conjugate, $M_{FB} = T_{1/2} M_V T_{1/2}^{-1}$ with the thermodynamic half step $T_{1/2}$: forward–backward is Verlet with the thermodynamic variables shifted by half a step. With the vertical part the same holds exactly with the transformation E_c, an explicit half step of the thermodynamic and vertical part (Section 6): EI($\frac{1}{2}$) and WRF($\frac{1}{2}$) are conjugate for the full linear system. Within a fast stage the conjugation acts only at the beginning and the end; at the stage boundaries of a MIS or RK3 step (where t and U start at the same time level) it leaves an error of first order in $\Delta\tau$ in the fast modes, which does not accumulate over the small steps. Verlet EI has no such defect.

7.2 Vertical: damping

For the vertical oscillator ($p' = -\omega W$, $W' = \omega p$, $a = \omega h$), both schemes are unconditionally stable; the squared modulus of the eigenvalues is the determinant of the step matrix:

$$\text{WRF } (t \text{ and } W \text{ off-centred}, \beta): \quad |\lambda|^2 = \frac{1 + (1-\beta)^2 a^2}{1 + \beta^2 a^2},$$

$$\text{EI } (W \text{ off-centred}, \gamma): \quad |\lambda|^2 = \frac{1 + (1-\gamma)\, a^2/2}{1 + \gamma\, a^2/2}.$$

Both are 1 for $\beta = \gamma = \frac{1}{2}$. For small a: $1 - (2\beta - 1)a^2$ against $1 - (2\gamma - 1)a^2/2$; for $a \to \infty$: $(1-\beta)^2/\beta^2$ against $(1-\gamma)/\gamma$. WRF's $\beta = 0.55$ gives $|\lambda| \to 0.82$, EI with $\gamma = 0.55$ only 0.90; the same limit needs $\gamma = 1/(1 + (1-\beta)^2/\beta^2) \approx 0.60$. Off-centring makes both schemes first order in $\Delta\tau$.

7.3 Summary

	Verlet EI	WRF forward–backward
horizontal integrator	Störmer–Verlet, symmetric	forward–backward (conjugate to Verlet)
order	2 ($\gamma = \frac{1}{2}$)	1 (2 in the staggered interpretation; $\beta = \frac{1}{2}$)
vertical integrator	trapezoidal in t, γ-weighted in W	β-weighted in t and W
damping	none ($\gamma = \frac{1}{2}$); W off-centring	off-centring β + divergence damping γ_d
unknown of the column solve	W (or Ω, Section 4)	W (WRF: after eliminating ϕ; Ω diagnostic)
metric term	explicit (or implicit with the Ω system)	explicit
evaluations per small step	each part once, 1 solve (half steps combined)	each part once, 1 solve
stage boundaries (MIS, RK3)	no defect	first-order conjugation defect

WRF needs its damping: RK3 has $|g_0| = 1$ for the acoustic modes (MIS study), and the forward–backward step adds none; divergence damping and off-centring provide it (Wicker and Skamarock 2002; Baldauf 2010). With MIS methods ($|g_0| < 1$) Verlet EI can be used without damping, and the off-centring γ is available where the vertically propagating acoustic modes have to be damped.

8 Implementation in the frozen split

Both schemes are fast integrators of the frozen MIS split, implemented in `frozen_fast_solve!` (`TimeStepping/frozen.jl`) with `fast = :ei` or `fast = :wrf`. They work for the three frozen models `FVModel` (full pressure `:gradp` or product form), `ExnerModel` and `MoistExnerModel`. In the frozen fast system $Z' = F_f(C, Z) + F$ the model splits F_f into the parts

`:hthermo`	H_t: $D_h U$ (incl. $-z_x \bar{U}$ in Ω), forcing of t and W, sponge
`:hmomentum`	H_U: horizontal pressure gradient $G_x p$ (incl. the metric term), forcing of U
`:vertical`	G_v: $D_v W$ in t, $G_z p - g\bar{\rho}$ in W

and provides the column solve $y = (I - DJ_v)^{-1} R$, J_v the Jacobian of G_v (exact for the linear vertical parts of the product, Exner and moist forms, the linearly implicit approximation for `:gradp`), D the row scaling c_t on t and c_w on W; the W system depends on $c_t c_w$ only and is factored once per stage (`frozen_stage_begin!`). Hooks (`FV/frozen_ei.jl`): `frozen_vertical_rhs!` (DG_v), `frozen_vertical_implicit!` ($Z \leftarrow Z + (I - DJ_v)^{-1} DG_v(Z)$) and `frozen_vertical_solve!` (general right-hand side).

Options. `frozen_fast_solve!`, `frozen_mis_step!` (MIS, Lawson, ETD) and `run_frozen` take the keywords `offcenter` $= \varepsilon$ and `smdiv` $= \gamma_d$ (default 0) for `:ei` and `:wrf` alike; `EI_GAMMA[]` (default $\frac{1}{2}$) selects version III of EI ($\varepsilon = 0$ only); `EI_MERGE[]` (default true) the combined half steps. The weights (e_t, e_w, i_t, i_w) are `ei_weights(h, ε)`.

Verlet EI (`:ei`):

$$
\begin{aligned}
&Z \leftarrow Z + e_t\, H_t(Z) + D_{(e_t,e_w)} G_v(Z) && E:\ t^{n+1/2},\ W \text{ explicit part}\\
&Z \leftarrow Z + h\, H_U\big(Z + \gamma_d (Z - Z_{\mathrm{prev}})\big) && K:\ U^{n+1}\\
&Z \leftarrow Z + i_t\, H_t(Z) && I:\ \text{horizontal part}\\
&Z \leftarrow Z + (I - D_{(i_t,i_w)} J_v)^{-1} D_{(i_t,i_w)} G_v(Z) && I:\ t^{n+1},\ W^{n+1};
\end{aligned}
$$

Z_{prev} the state of the previous kick, $Z_{\mathrm{prev}} = Z$ after the first E. For linear G_v and $\varepsilon = 0$, $\gamma = \frac{1}{2}$ this is exactly (1)–(4). Combined half steps as in Section 6.3: one horizontal momentum, one horizontal thermodynamic and one vertical evaluation and one column solve per small step — as WRF; only the first E of a stage is extra. For `:gradp` (linearly implicit) and in the sponge (evaluated at the state before I) combined and separate half steps differ by $O(\Delta\tau^2)$ per step ($\le 3 \cdot 10^{-6}$ relative in the test), and both are of order 2 for $\varepsilon = 0$.

WRF (`:wrf`):

$$Z \leftarrow Z + h\, H_U\big(Z + \gamma_d (Z - Z_{\mathrm{prev}})\big) \qquad K:\ U^{n+1}$$
$$Z \leftarrow Z + (I - \beta h J_v)^{-1} h \big[H_t(Z) + G_v(Z)\big] \qquad I_{\beta h} E_{(1-\beta)h}:\ t^{n+1},\ W^{n+1},$$

$Z_{\mathrm{prev}} = Z$ at the start of the stage. For linear G_v the second line is $Y^{n+1} = Y^n + h[H_t + (1 - \beta)A_v Y^n + \beta A_v Y^{n+1}]$ of Section 5.

Tests (`test/runtests.jl`, "Verlet EI and WRF fast integrators"): the two-factor column solve is exact for the linear forms; on one frozen stage of the hill problem (product, `:gradp`, Exner, pressure, moist) EI is of order 2 and WRF of order 1 against RK4, both of order 1 with $\varepsilon = 0.1$; $\mathrm{WRF}^n E = E\,\mathrm{EI}^n$ for $(\varepsilon, \gamma_d) = (0,0), (0.1,0), (0.1,0.1)$; on a flat, horizontally uniform column EI and WRF with $\varepsilon = 0$ coincide with the Crank–Nicolson step of `:hevi` to round-off, and EI stays bounded at the vertical Courant number 35; in MIS2 both conserve mass to round-off on terrain, EI differs from `:hevi` by $O(\Delta\tau^2)$, WRF by $O(\Delta\tau)$; with $\gamma = 0.6$ (version III) EI is of order 1. Combined and separate half steps of EI agree to round-off for the linear forms ($\gamma = 0.5$ and 0.6); no small step is longer than $\Delta t / n_{\mathrm{sub}}$ (Section 9). Full suite: 571 tests pass.

error on one frozen stage ($T = 8$ s, max norm)	8 steps	16 steps	32 steps
EI	0.0242	0.0060	0.0015
WRF, $\beta = 0.5$	0.116	0.055	0.027
WRF, $\beta = 0.55$	0.117	0.057	0.028
WRF, $\beta = 0.55$, $\gamma_d = 0.1$	0.102	0.050	0.025

(Exner form; the other forms agree to two digits.) The linearized path (`fast_solve!`, `:ei`, `:wrf` in `TimeStepping/fast.jl`) has the same two loops for the linear fast operator.

9 Agnesi mountain wave

Set-up as in the MIS study (`docs/mis_agnesi`): hydrostatic wave over the Agnesi hill ($a = 10$ km, $h_0 = 400$ m, $U = 10$ m/s, $N = 0.01\,\mathrm{s}^{-1}$), 200×78 cells, $\Delta x = 2$ km, $\Delta\eta = 200$ m, $T = 10800$ s, $n_{\mathrm{sub}} = \lceil \Delta t / \Delta\tau_{\mathrm{ac}} \rceil$ with $\Delta\tau_{\mathrm{ac}} = 4.3$ s (3 small steps at $\Delta t = 10$ s, 56 at 240 s); reference RK4 of the same semi-discrete operator; errors are relative L^2 errors in $|x - x_m| \le 30$ km, $z \le 9$ km. Two forms of the extended system (`ExnerModel`) with an additional pressure variable: the *Exner pressure* π (W row $-c_p \Theta_f \delta\pi/\Delta z - g\bar{\rho}$) and the *pressure* p (W row $-\delta p/\Delta z - g\bar{\rho}$, `pvar = :p`); each column is in discrete hydrostatic balance of its form (rest-state residual of the W rows $4 \cdot 10^{-13}$). HEVI is the fast integrator of the MIS study (horizontal Verlet half step, Crank–Nicolson vertical step, horizontal half step). EI and WRF are run with the same options: none, $\varepsilon = 0.1$, and $\varepsilon = \gamma_d = 0.1$ (WRF's standard values).

`FORM=exner|pres FAST=ei|wrf|hevi [OFFCENTER=`ε`] [SMDIV=`γ_d`] julia examples/frozen_agnesi.jl`

Table 1: Exner pressure: relative L^2 error of θ' against Δt (s). u: unstable; bold: error ≥ 0.1.

	fast integrator	10	20	40	60	90	120	180	240
MIS2	HEVI	0.007	0.015	0.015	0.018	0.034	0.038	**0.132**	u
	EI	0.002	0.002	0.004	0.005	0.026	0.034	**0.126**	u
	WRF	0.003	0.002	0.004	0.005	0.030	0.044	**0.315**	u
	EI, $\varepsilon = 0.1$	0.002	0.002	0.003	0.004	0.011	0.031	**0.164**	u
	WRF, $\varepsilon = 0.1$	0.002	0.002	0.003	0.003	0.014	0.038	**0.267**	u
	EI, $\varepsilon = \gamma_d = 0.1$	0.001	0.002	0.003	0.004	0.009	0.017	0.036	u
	WRF, $\varepsilon = \gamma_d = 0.1$	0.002	0.002	0.003	0.003	0.010	0.020	0.046	u
MIS3C	HEVI	0.005	0.018	0.035	u	u	u	u	u
	EI	0.002	0.002	0.031	u	u	u	u	u
	WRF	0.002	0.003	0.029	u	u	u	u	u
	EI, $\varepsilon = 0.1$	0.002	0.002	0.002	0.004	0.006	**0.846**	u	u
	WRF, $\varepsilon = 0.1$	0.002	0.002	0.003	0.005	0.007	**1.340**	u	u
	EI, $\varepsilon = \gamma_d = 0.1$	0.002	0.002	0.002	0.004	0.005	0.008	**0.176**	**0.261**
	WRF, $\varepsilon = \gamma_d = 0.1$	0.002	0.002	0.002	0.004	0.006	0.008	**0.196**	**0.289**
MIS4a	HEVI	0.005	0.015	0.015	0.090	u	**0.153**	u	u
	EI	0.001	0.002	0.004	**0.141**	u	**1.050**	u	u
	WRF	**1.186**	0.002	0.004	u	u	u	u	u
	EI, $\varepsilon = 0.1$	0.002	0.002	0.003	0.005	0.011	0.010	0.026	**2.037**
	WRF, $\varepsilon = 0.1$	**0.413**	0.002	0.003	0.005	0.011	0.010	0.030	u
	EI, $\varepsilon = \gamma_d = 0.1$	0.001	0.002	0.003	0.004	0.009	0.009	0.021	0.032
	WRF, $\varepsilon = \gamma_d = 0.1$	0.002	0.002	0.003	0.005	0.010	0.009	0.023	0.032
WSRK3	HEVI	u	u	u	u	u	u	u	u
	EI	0.013	u	u	u	u	u	u	u
	WRF	u	u	u	u	u	u	u	u
	EI, $\varepsilon = 0.1$	0.002	0.002	u	u	**0.135**	u	u	**0.672**
	WRF, $\varepsilon = 0.1$	0.002	0.002	u	u	**0.191**	u	u	**0.719**
	EI, $\varepsilon = \gamma_d = 0.1$	0.001	0.002	u	u	0.013	0.069	**0.131**	**0.172**
	WRF, $\varepsilon = \gamma_d = 0.1$	0.002	0.002	u	u	0.013	0.080	**0.145**	**0.183**

Number of small steps per stage (corrected). Stage i of a MIS step integrates over $d_i \Delta t$. The number of small steps was $\max(1, \mathrm{round}(d_i n_{\mathrm{sub}}))$, which allows small steps up to $1.5\,\Delta t/n_{\mathrm{sub}}$, i.e. above the horizontal acoustic limit for which n_{sub} was chosen. MIS2 ($d = (0.127, 0.590, 0.470)$) with $n_{\mathrm{sub}} = 3$ at $\Delta t = 10\,\mathrm{s}$ ran its third stage in one small step of $4.7\,\mathrm{s} > \Delta\tau_{\mathrm{ac}} = 4.3\,\mathrm{s}$, and with $n_{\mathrm{sub}} = 5$ at 20 s in two steps of 4.7 s: exactly the two cases in which MIS2 with EI and WRF was unstable in the first version of this note. HEVI survived them, because its horizontal Verlet steps are half steps. The count is now

$$n_i = \max\left(1, \lceil d_i n_{\mathrm{sub}} \rceil\right) \qquad (\texttt{fast_substeps}),$$

so that no small step is longer than $\Delta t/n_{\mathrm{sub}}$, for MIS, ETD, Lawson and EPIRK and in the frozen and the linearized split. The MIS study of `docs/mis_agnesi` used the rounded count; its HEVI numbers change slightly (e.g. MIS2 at 10 s: θ' error 0.014 → 0.007). All results below use the corrected count.

EI and WRF with the same options behave the same. In both forms and for all three option sets the stable and unstable cases of EI and WRF coincide, and the errors agree to one or two digits, EI's being slightly smaller (no change of variables missing at the stage boundaries). The few differences are at the edges, where the stage-boundary defect of WRF matters: MIS4a at 10 s ($n_{\mathrm{sub}} = 3$, first stage one step) is weakly unstable with WRF for $\gamma_d = 0$ (Exner: θ' error 1.19 and 0.41; pressure: 0.50 and 0.18) and fine with EI; WSRK3 at 10 and 20 s without options (w error 7.2 and 4.6 with WRF against 0.17 and 0.15 with EI, pressure form).

Table 2: Pressure as additional variable: relative L^2 error of θ' against Δt (s).

	fast integrator	10	20	40	60	90	120	180	240
MIS2	HEVI	0.008	0.015	0.014	0.019	0.023	0.018	0.040	u
	EI	0.003	0.002	0.006	0.005	0.019	0.010	0.033	u
	WRF	0.003	0.003	0.006	0.005	0.019	0.011	0.032	u
	EI, $\varepsilon = 0.1$	0.002	0.002	0.004	0.004	0.008	0.008	0.011	u
	WRF, $\varepsilon = 0.1$	0.002	0.002	0.004	0.004	0.007	0.009	0.011	u
	EI, $\varepsilon = \gamma_d = 0.1$	0.001	0.002	0.004	0.004	0.007	0.008	0.010	u
	WRF, $\varepsilon = \gamma_d = 0.1$	0.002	0.002	0.004	0.004	0.007	0.008	0.010	u
MIS3C	HEVI	0.005	0.018	0.045	0.027	0.032	0.061	0.058	u
	EI	0.002	0.002	0.042	0.029	0.038	0.070	0.055	0.090
	WRF	0.002	0.002	0.039	0.029	0.037	0.068	0.054	0.088
	EI, $\varepsilon = 0.1$	0.002	0.002	0.002	0.003	0.003	0.004	0.006	0.013
	WRF, $\varepsilon = 0.1$	0.002	0.002	0.002	0.003	0.003	0.004	0.006	0.013
	EI, $\varepsilon = \gamma_d = 0.1$	0.002	0.002	0.002	0.002	0.003	0.003	0.005	0.013
	WRF, $\varepsilon = \gamma_d = 0.1$	0.002	0.002	0.002	0.002	0.003	0.003	0.004	0.013
MIS4a	HEVI	0.005	0.015	0.015	0.089	u	0.032	u	u
	EI	0.001	0.002	0.003	**0.156**	u	**0.102**	u	u
	WRF	**0.500**	0.002	0.004	u	u	u	u	u
	EI, $\varepsilon = 0.1$	0.002	0.002	0.002	0.003	0.003	0.006	0.005	0.007
	WRF, $\varepsilon = 0.1$	**0.184**	0.002	0.003	0.003	0.004	0.007	0.005	0.007
	EI, $\varepsilon = \gamma_d = 0.1$	0.001	0.002	0.002	0.002	0.003	0.006	0.005	0.006
	WRF, $\varepsilon = \gamma_d = 0.1$	0.002	0.002	0.002	0.002	0.003	0.006	0.004	0.006
WSRK3	HEVI	u	u	0.040	0.047	0.059	0.054	0.090	0.099
	EI	0.008	0.032	0.050	0.052	0.073	0.068	**0.102**	**0.111**
	WRF	0.021	0.052	0.050	0.052	0.077	0.068	**0.104**	**0.113**
	EI, $\varepsilon = 0.1$	0.002	0.002	0.002	0.004	0.007	0.005	0.015	0.016
	WRF, $\varepsilon = 0.1$	0.002	0.002	0.003	0.004	0.007	0.005	0.015	0.017
	EI, $\varepsilon = \gamma_d = 0.1$	0.001	0.002	0.002	0.003	0.006	0.004	0.010	0.016
	WRF, $\varepsilon = \gamma_d = 0.1$	0.002	0.002	0.003	0.004	0.006	0.004	0.010	0.016

Accuracy. With $\varepsilon = 0.1$, EI and WRF reduce the θ' error at $\Delta t \le 60\,\mathrm{s}$ from 0.5–$4.5 \cdot 10^{-2}$ (HEVI) to 1.4–$5 \cdot 10^{-3}$, for all four methods and both forms (without options at $\Delta t \le 20\,\mathrm{s}$): with HEVI the θ' error is dominated by the horizontal–vertical splitting of the fast step (the Strang splitting of the stiff vertical part), not by the MIS method. EI and WRF use $D_h U^{n+1}$ and $D_v W^{n+1}$ in the same thermodynamic update and avoid it. The w error (≈ 0.09) is the same for all stable runs: the undamped vertical acoustic modes of the MIS study.

Off-centring and extrapolation.

- Without options EI, WRF and HEVI have irregular stability gaps in the Exner form (MIS3C from 60 s, MIS4a at 60–120 s, WSRK3 almost everywhere); in the pressure form MIS2 (up to 180 s), MIS3C and WSRK3 are stable (HEVI: WSRK3 unstable at 10–20 s), MIS4a still has the gap at 60–90 s and from 180 s (WRF from 60 s).
- $\varepsilon = 0.1$ removes the gaps at small and medium Δt and reduces the errors at $\Delta t \ge 40\,\mathrm{s}$ by a factor 2–20. In the pressure form all four methods are then stable up to 240 s (MIS2 up to 180 s), WSRK3 included, with θ' errors ≤ 0.017.
- $\gamma_d = 0.1$ matters only in the Exner form at $\Delta t \ge 120\,\mathrm{s}$ (MIS3C, MIS4a and WSRK3 stable up to 240 s except WSRK3 at 40–60 s); in the pressure form it changes the errors by a few per cent.

Pressure against Exner pressure. With the pressure as additional variable all methods are markedly more robust: with $\varepsilon = 0.1$, MIS3C at 240 s has θ' error 0.013 (Exner: unstable; 0.26

Table 3: Exner pressure: relative L^2 error of w. Bold: error ≥ 1.

	fast integrator	10	20	40	60	90	120	180	240
MIS2	HEVI	0.056	0.104	0.100	0.107	0.126	0.205	**1.173**	u
	EI	0.054	0.098	0.093	0.097	0.125	0.203	**1.211**	u
	WRF	0.068	0.100	0.095	0.099	0.147	0.262	**2.503**	u
	EI, $\varepsilon = 0.1$	0.089	0.095	0.093	0.096	0.092	0.191	**1.341**	u
	WRF, $\varepsilon = 0.1$	0.089	0.096	0.094	0.098	0.106	0.237	**2.088**	u
	EI, $\varepsilon = \gamma_d = 0.1$	0.091	0.092	0.092	0.095	0.087	0.154	0.350	u
	WRF, $\varepsilon = \gamma_d = 0.1$	0.092	0.093	0.094	0.096	0.090	0.172	0.408	u
MIS3C	HEVI	0.186	0.139	0.139	u	u	u	u	u
	EI	0.186	0.136	0.134	u	u	u	u	u
	WRF	0.193	0.150	0.121	u	u	u	u	u
	EI, $\varepsilon = 0.1$	0.089	0.090	0.088	0.164	0.087	**6.007**	u	u
	WRF, $\varepsilon = 0.1$	0.089	0.091	0.089	0.180	0.089	**9.494**	u	u
	EI, $\varepsilon = \gamma_d = 0.1$	0.091	0.091	0.090	0.102	0.087	0.106	**1.133**	**1.800**
	WRF, $\varepsilon = \gamma_d = 0.1$	0.092	0.092	0.090	0.104	0.088	0.107	**1.256**	**1.972**
MIS4a	HEVI	0.154	0.093	0.097	**7.068**	u	**1.892**	u	u
	EI	0.151	0.087	0.090	**8.905**	u	**9.743**	u	u
	WRF	**1.922**	0.089	0.090	u	u	u	u	u
	EI, $\varepsilon = 0.1$	0.090	0.091	0.091	0.095	0.150	0.092	0.173	**14.300**
	WRF, $\varepsilon = 0.1$	**8.840**	0.093	0.092	0.097	0.152	0.092	0.191	u
	EI, $\varepsilon = \gamma_d = 0.1$	0.091	0.091	0.091	0.096	0.145	0.086	0.151	0.085
	WRF, $\varepsilon = \gamma_d = 0.1$	0.092	0.092	0.091	0.096	0.146	0.086	0.154	0.090
WSRK3	HEVI	u	u	u	u	u	u	u	u
	EI	**3.592**	u	u	u	u	u	u	u
	WRF	u	u	u	u	u	u	u	u
	EI, $\varepsilon = 0.1$	0.093	0.092	u	u	0.934	u	u	**2.555**
	WRF, $\varepsilon = 0.1$	0.094	0.093	u	u	**1.318**	u	u	**2.780**
	EI, $\varepsilon = \gamma_d = 0.1$	0.091	0.091	u	u	0.178	**2.278**	0.522	0.589
	WRF, $\varepsilon = \gamma_d = 0.1$	0.092	0.092	u	u	0.185	**2.526**	0.620	0.631

with $\gamma_d = 0.1$), MIS4a 0.0065 (Exner: 2.0 or unstable; 0.032 with γ_d), WSRK3 0.016 (Exner with γ_d: unstable at 40–60 s, 0.17 at 240 s), and HEVI makes WSRK3 stable from 40 s. A plausible cause, not analysed here: in the pressure form the W row $-\delta p$ has no frozen coefficient, while the Exner form carries the frozen $c_p\Theta_f$ in front of $\delta\pi$ and in the π equation, so that the frozen fast operator of a stage and the slow remainder differ more (the constraint-defect modes of the MIS study).

Cost. Serial runs, MIS4a at $\Delta t = 60$ s ($n_{\text{sub}} = 14$, $T = 10800$ s, the first Δt of each run excluded because of compilation; Exner form): HEVI 89 s, EI (combined half steps, version III $\gamma = 0.55$) 49 s, WRF ($\varepsilon = \gamma_d = 0.1$) 44 s. The combined EI costs 10 % more than WRF (the first E of each stage and, for version III, the column-local correction) and 55 % of HEVI.

10 Open decisions

1. Default fast integrator: EI or WRF, both with $\varepsilon = 0.1$ (and $\gamma_d = 0.1$ for the Exner form). They are conjugate; EI avoids the stage-boundary defect at the same cost.

2. Default pressure variable of the extended system: p is markedly more robust than π on Agnesi; to be checked on the moist cases (`MoistExnerModel`, `pvar = :p`).

3. The remaining gaps (Exner form; MIS4a without options): linear stability analysis of the coupled MIS–EI/WRF step.

4. Rerun of the MIS study (`docs/mis_agnesi`) with the corrected number of small steps.

Table 4: Pressure as additional variable: relative L^2 error of w.

	fast integrator	10	20	40	60	90	120	180	240
MIS2	HEVI	0.074	0.099	0.096	0.104	0.113	0.111	0.108	u
	EI	0.077	0.091	0.091	0.095	0.105	0.103	0.099	u
	WRF	0.094	0.092	0.093	0.097	0.107	0.105	0.102	u
	EI, $\varepsilon = 0.1$	0.088	0.095	0.091	0.095	0.098	0.102	0.098	u
	WRF, $\varepsilon = 0.1$	0.088	0.095	0.092	0.097	0.100	0.104	0.101	u
	EI, $\varepsilon = \gamma_d = 0.1$	0.091	0.091	0.091	0.093	0.097	0.101	0.096	u
	WRF, $\varepsilon = \gamma_d = 0.1$	0.091	0.092	0.093	0.095	0.099	0.103	0.099	u
MIS3C	HEVI	0.189	0.156	0.122	0.104	0.112	0.130	0.148	u
	EI	0.187	0.149	0.110	0.096	0.102	0.125	0.117	0.141
	WRF	0.195	0.166	0.110	0.098	0.103	0.123	0.116	0.137
	EI, $\varepsilon = 0.1$	0.089	0.089	0.093	0.089	0.097	0.092	0.093	0.086
	WRF, $\varepsilon = 0.1$	0.089	0.089	0.094	0.090	0.098	0.093	0.093	0.087
	EI, $\varepsilon = \gamma_d = 0.1$	0.091	0.091	0.091	0.090	0.095	0.091	0.091	0.087
	WRF, $\varepsilon = \gamma_d = 0.1$	0.092	0.092	0.092	0.091	0.095	0.092	0.091	0.088
MIS4a	HEVI	0.141	0.106	0.097	**5.841**	u	0.883	u	u
	EI	0.137	0.102	0.091	**9.667**	u	**1.184**	u	u
	WRF	**11.220**	0.106	0.092	u	u	u	u	u
	EI, $\varepsilon = 0.1$	0.090	0.092	0.091	0.089	0.095	0.098	0.086	0.094
	WRF, $\varepsilon = 0.1$	**7.884**	0.093	0.092	0.090	0.096	0.100	0.086	0.095
	EI, $\varepsilon = \gamma_d = 0.1$	0.091	0.091	0.090	0.090	0.094	0.097	0.087	0.093
	WRF, $\varepsilon = \gamma_d = 0.1$	0.092	0.092	0.091	0.090	0.096	0.099	0.087	0.094
WSRK3	HEVI	u	u	0.143	0.159	0.204	0.128	0.133	0.162
	EI	0.174	0.149	0.146	0.139	0.223	0.098	0.154	0.140
	WRF	**7.230**	**4.551**	0.144	0.145	0.234	0.114	0.159	0.141
	EI, $\varepsilon = 0.1$	0.093	0.094	0.089	0.087	0.087	0.080	0.092	0.080
	WRF, $\varepsilon = 0.1$	0.094	0.095	0.090	0.088	0.087	0.080	0.092	0.077
	EI, $\varepsilon = \gamma_d = 0.1$	0.091	0.090	0.089	0.087	0.082	0.082	0.085	0.083
	WRF, $\varepsilon = \gamma_d = 0.1$	0.092	0.091	0.090	0.089	0.082	0.081	0.085	0.082

5. W system with explicit metric term (sufficient for $c_s\Delta\tau|z_x|/(J\Delta\eta) \lesssim 1$) or the Ω system with implicit metric term; for the Ω system the column-local approximations (a), (b) or the exact two-dimensional system.

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